

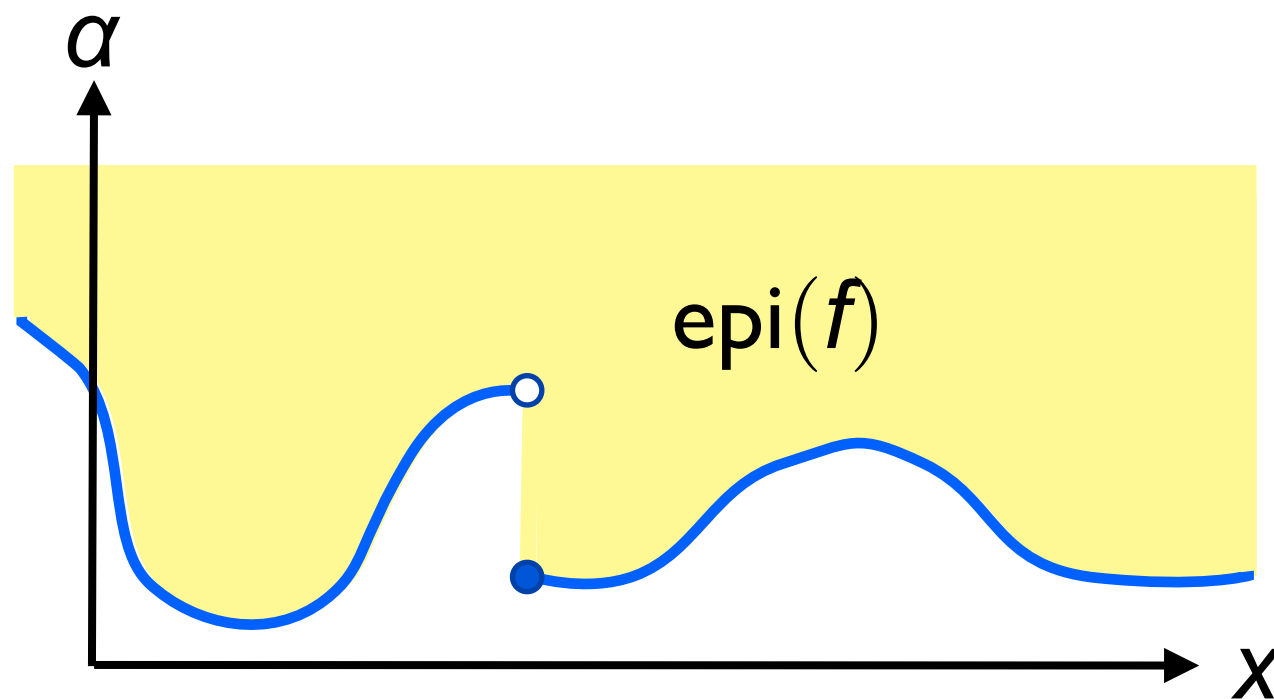
MGT-418 Convex Optimization: Convex Functions

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Epigraph and Domain

Definition: The *epigraph* of $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is the set

$$\text{epi}(f) = \{(x, \alpha) \in \mathbb{R}^{n+1} : \alpha \geq f(x)\}$$



Definition: The *domain* of $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is the set

$$\text{dom}(f) = \{x \in \mathbb{R}^n : f(x) < +\infty\}$$

Definition: $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is called *proper* if $\text{dom}(f) \neq \emptyset$.

Convex Functions

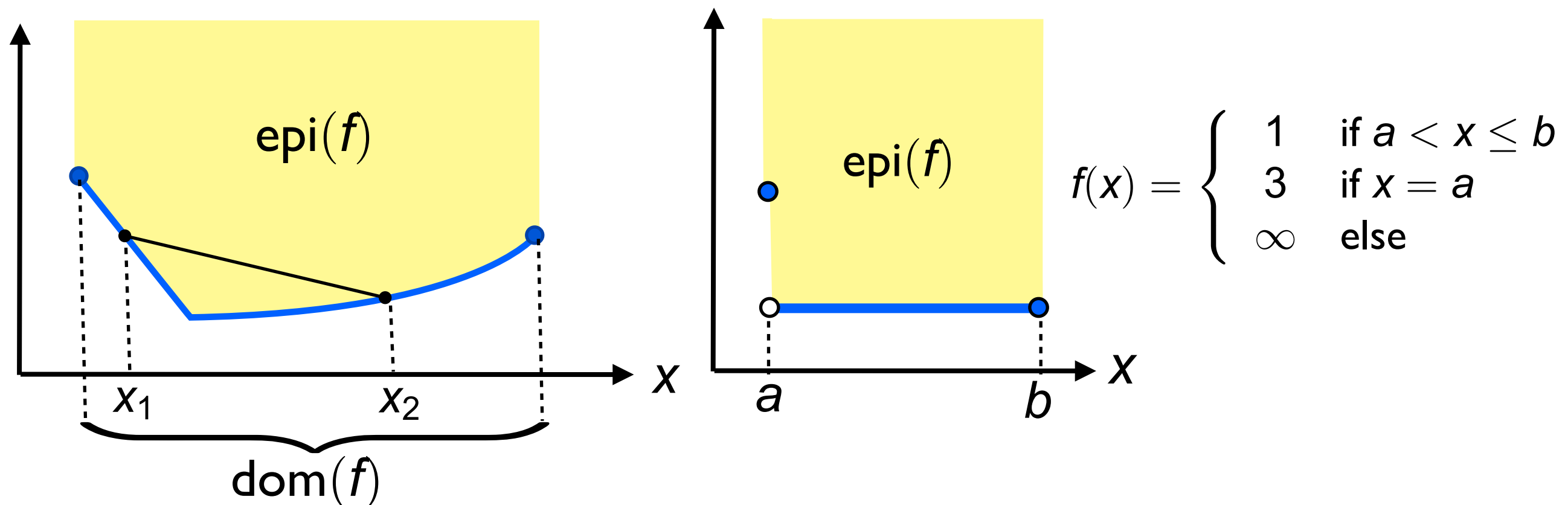
Definition: A function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *convex* if its epigraph is a convex set.

f is convex if and only if its *domain* is a *convex* set and

$$f(\theta x_1 + (1 - \theta)x_2) \leq \theta f(x_1) + (1 - \theta)f(x_2)$$

for all $x_1, x_2 \in \text{dom}(f)$, $\theta \in [0, 1]$.

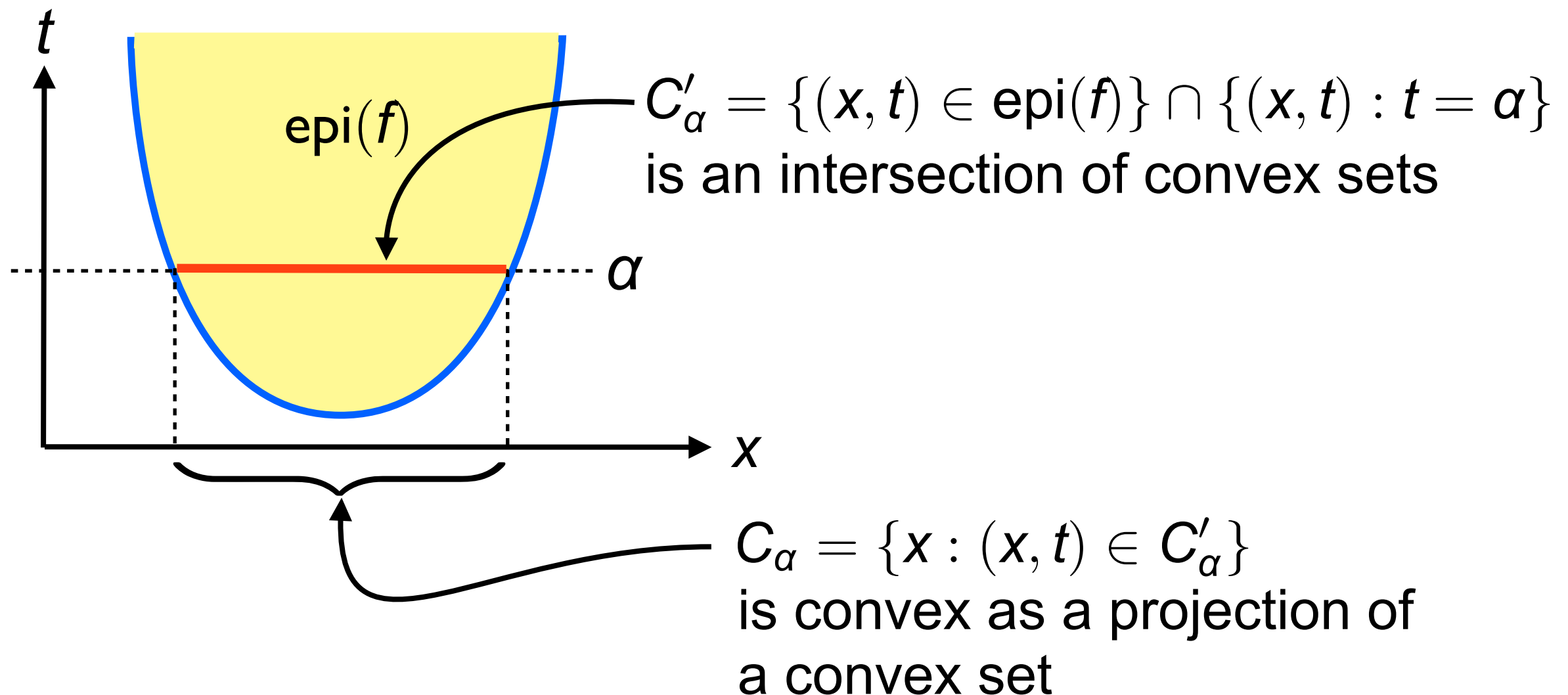
Examples:



Sublevel Sets

Definition: The α -*sublevel set* of a function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is defined as $C_\alpha = \{x : f(x) \leq \alpha\}$.

Proposition: If $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is convex, then all of its sublevel sets are convex. The reverse implication is *not* true



Examples of Convex Functions

Univariate functions:

		Domain
• Exponential functions	$f(x) = e^{ax}$	$\mathbb{R}$
• Powers	$f(x) = x^\alpha$ ($\alpha \geq 1$ or $\alpha \leq 0$)	$\mathbb{R}_{++}$
• Negative logarithm	$f(x) = -\log(x)$	$\mathbb{R}_{++}$
• Negative entropy	$f(x) = x \log(x)$	$\mathbb{R}_{++}$

Multivariate functions:

• Affine functions	$f(x) = a^\top x + b$	$\mathbb{R}^n$
• p -Norms ($p \geq 1$)	$f(x) = \ x\ _p = (\sum_{i=1}^n x_i ^p)^{1/p}$	$\mathbb{R}^n$
• ∞ -Norm	$f(x) = \ x\ _\infty = \max_i x_i $	$\mathbb{R}^n$
• Indicator function of a convex set C	$f(x) = \begin{cases} 0 & \text{if } x \in C \\ \infty & \text{else} \end{cases}$	C

*Convention: $\mathbb{R}_{++}^n = \{x \in \mathbb{R}^n : x_i > 0 \forall i = 1, \dots, n\}$.

Examples of Convex Functions

Matrix functions:

Domain

- Trace functions (i.e., linear functions)

$$f(X) = \text{tr}(A^\top X) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} X_{ij} \quad (A \in \mathbb{R}^{m \times n})$$

$\mathbb{R}^{m \times n}$

- Maximum eigenvalue $f(X) = \lambda_{\max}(X)$

$\mathbb{S}^n$

- Spectral norm

$$f(X) = \|X\|_2 = \sup_{v \neq 0} \|Xv\|_2 / \|v\|_2$$

$\mathbb{R}^{m \times n}$

(maximum scale by which X can stretch a vector)

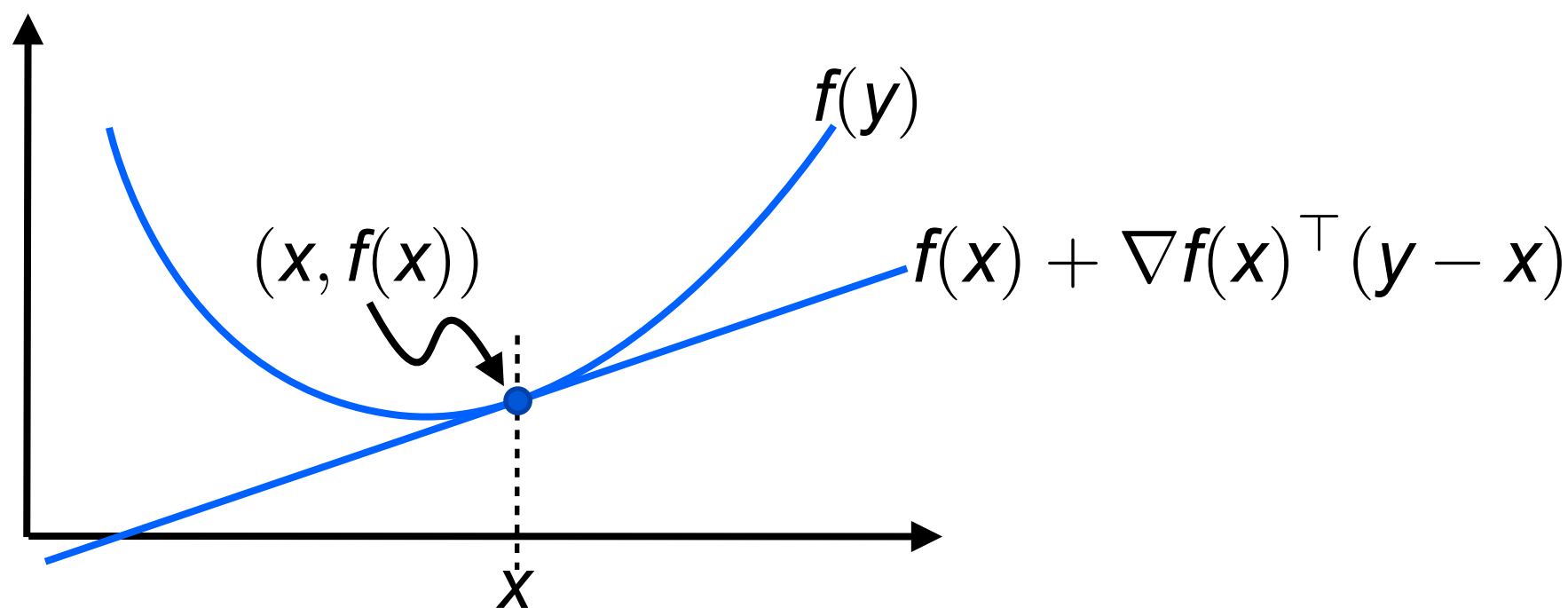
1st-Order Conditions

Definition: A function $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *differentiable* if its gradient $\nabla f = (\partial f / \partial x_1, \dots, \partial f / \partial x_n)^\top$ exists at each point in $\text{dom}(f)$, and $\text{dom}(f)$ is open.

Proposition: A *differentiable function* $f: \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *convex* if and only if $\text{dom}(f)$ is convex and

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \text{dom}(f).$$

$\Rightarrow$ *1st-order Taylor approximation* underestimates f *globally*.



$\Rightarrow$ From *local information* about a convex function we can obtain *global information*.

Univariate Functions*

Proposition: A *differentiable function* $f : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* if and only if $f(y) \geq f(x) + f'(x)(y - x) \quad \forall x, y \in \mathbb{R}$.

Proof:

$\implies$: If $x, y \in \mathbb{R}$, $0 < t \leq 1$ then

$$\begin{aligned} f(x + t(y - x)) &\leq (1 - t)f(x) + tf(y) && \text{(convexity)} \\ f(y) - f(x) &\geq [f(x + t(y - x)) - f(x)]/t && \text{(divide by } t) \\ f(y) - f(x) &\geq f'(x)(y - x) && \text{(limit } t \downarrow 0) \end{aligned}$$

$\Longleftarrow$: For $x, y \in \mathbb{R}$, $0 \leq t \leq 1$, set $z = tx + (1 - t)y$.

$$\begin{aligned} t[f(x) - f(z)] &\geq tf'(z)(x - z) && \text{(by assumption)} \\ (1 - t)[f(y) - f(z)] &\geq (1 - t)f'(z)(y - z) && \text{(by assumption)} \\ tf(x) + (1 - t)f(y) &\geq f(z) && \text{(sum of above)} \end{aligned}$$

1st-Order Conditions*

Proposition: A *differentiable function* $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* if and only if $f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in \mathbb{R}^n$.

Proof:

$\implies$: $g(t) = f(ty + (1 - t)x)$ is convex in t for any $x, y \in \mathbb{R}^n$.

$$\begin{aligned} g'(t) &= \nabla f(ty + (1 - t)x)^\top (y - x) && \text{(definition of } g) \\ g(1) &\geq g(0) + g'(0) && \text{(univariate case)} \\ f(y) &\geq f(x) + \nabla f(x)^\top (y - x) && \text{(substitution)} \end{aligned}$$

$\Longleftarrow$: $x, y \in \mathbb{R}^n$, $t, \tilde{t} \in \mathbb{R}$, $z = ty + (1 - t)x$, $\tilde{z} = \tilde{t}y + (1 - \tilde{t})x$

$$\begin{aligned} f(z) &\geq f(\tilde{z}) + \nabla f(\tilde{z})^\top (z - \tilde{z}) && \text{(by assumption)} \\ g(t) &\geq g(\tilde{t}) + g'(\tilde{t})(t - \tilde{t}) && \text{(definition of } g, z, \tilde{z}) \end{aligned}$$

By the 1st-order condition for univariate functions, g is convex. Thus, f is also convex.

2nd-Order Conditions

Definition: A function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *twice differentiable* if its Hessian

$$\nabla^2 f = \begin{pmatrix} \partial^2 f / \partial x_1 \partial x_1 & \cdots & \partial^2 f / \partial x_1 \partial x_n \\ \vdots & \ddots & \vdots \\ \partial^2 f / \partial x_n \partial x_1 & \cdots & \partial^2 f / \partial x_n \partial x_n \end{pmatrix}$$

exists at each point in $\text{dom}(f)$, and $\text{dom}(f)$ is open.

Proposition: A *twice differentiable function* $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *convex* if and only if $\text{dom}(f)$ is convex and

$$\nabla^2 f(x) \succeq 0 \quad \forall x \in \text{dom}(f).$$

The condition $\nabla^2 f(x) \succeq 0$ can be interpreted geometrically as the requirement that f has *upward curvature* at x .

Univariate Functions*

Proposition: A *twice differentiable function* $f : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* if and only if $f''(x) \geq 0 \ \forall x \in \mathbb{R}$.

Proof:

$\implies$: If $x, y \in \mathbb{R}$, $y > x$, then

$$\begin{aligned} f(y) &\geq f(x) + f'(x)(y - x) && \text{(1st-order cond.)} \\ f(x) &\geq f(y) + f'(y)(x - y) && \text{(1st-order cond.)} \\ 0 &\geq [f'(x) - f'(y)]/(y - x) && \text{(sum of above } \times (y - x)^{-2}) \\ 0 &\leq f''(x) && \text{(limit } y \downarrow x) \end{aligned}$$

$\Leftarrow$: For $x, y \in \mathbb{R}$ we have

$$\begin{aligned} f(y) &= f(x) + \int_x^y f'(u) du \\ &= f(x) + \int_x^y f'(x) + \int_x^u f''(v) dv du \\ &\geq f(x) + \int_x^y f'(x) du \\ &= f(x) + f'(x)(y - x) \end{aligned}$$

Thus, f is convex as it satisfies the 1st-order condition.

2nd-Order Conditions*

Proposition: A *twice differentiable function* $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* if and only if $\nabla^2 f(x) \succeq 0 \ \forall x \in \mathbb{R}^n$.

Proof:

$\implies$: $g(t) = f(x + ty)$ is convex in t for any $x, y \in \mathbb{R}^n$.

$$\begin{aligned} g''(t) &= y^\top \nabla^2 f(x + ty) y && \text{(definition of } g) \\ g''(0) &\geq 0 && \text{(univariate case)} \\ \nabla^2 f(x) &\succeq 0 && \text{(as } y \text{ is arbitrary)} \end{aligned}$$

$\impliedby$: Define g as above. Then, we have for any t

$$\begin{aligned} \nabla^2 f(x + ty) &\succeq 0 && \text{(by assumption)} \\ g''(t) &\geq 0 && \text{(definition of } g) \end{aligned}$$

By the 2nd-order condition for univariate functions, g is convex. Thus, f is also convex.

Examples

- *Quadratic functions* $f(x) = \frac{1}{2}x^\top Px + q^\top x + r$
are convex if $\nabla^2 f(x) = P \succeq 0$.
- The *least-squares objective* $f(x) = \|Ax - b\|_2^2$ is convex because $\nabla^2 f(x) = 2A^\top A \succeq 0$ for all $A \in \mathbb{R}^{m \times n}$.
- *Quadratic-over-linear* functions of the form $f(x, y) = x^2/y$ are convex as long as $y > 0$ because

$$\nabla^2 f(x, y) = \frac{2}{y^3} \begin{pmatrix} y \\ -x \end{pmatrix} \begin{pmatrix} y \\ -x \end{pmatrix}^\top \succeq 0 \text{ for } y > 0.$$

Checking Convexity along Lines

A function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ is *convex* if and only if each *univariate* function $g : \mathbb{R} \rightarrow (-\infty, +\infty]$ of the form

$$g(t) = f(a + tb) \text{ for } a, b \in \mathbb{R}^n$$

is convex in t .

$\implies$ One can check convexity of a multivariate function by restricting it to a line.

Negative Log-Determinant

Proposition: The *log-determinant function* $f(X) = -\log \det(X)$ is *convex* on the set of positive definite matrices $\mathbb{S}_{++}^n$.

Proof:

Show that $g(t) = -\log \det(X + tV)$ is convex $\forall X \in \mathbb{S}_{++}^n, V \in \mathbb{S}^n$. Thus,

$$\begin{aligned} g(t) &= -\log \det(X^{1/2} [I + tX^{-1/2} V X^{-1/2}] X^{1/2}) \\ &= -\log \det(X^{1/2} R [I + tD] R^\top X^{1/2}), \end{aligned}$$

where $X^{-1/2} V X^{-1/2} = R D R^\top$ for R orthogonal ($R^\top = R^{-1}$) and $D = \text{diag}(d_1, \dots, d_n)$. As $\det(AB) = \det(BA) = \det(A)\det(B)$, we have

$$\begin{aligned} g(t) &= -\log \det([R^\top X R][I + tD]) = -\log \det(X) - \log \det(I + tD) \\ &= -\log \det(X) - \sum_{i=1}^n \log(1 + t d_i), \end{aligned}$$

which is manifestly convex in t .

Convexity-Preserving Transformations

Sometimes one can establish convexity of f by showing that f is obtained from simple convex functions via *transformations that preserve convexity*:

- non-negative weighted sum
- composition with affine function
- pointwise maximum and supremum
- composition
- minimization
- perspective

Affine Transformations

Affine transformation of *inputs*: If f is convex, then $g(x) = f(Ax + b)$ is also convex.

Non-negative affine transformation of *outputs*: If $f_1, \dots, f_K$ are convex functions and $\rho_1, \dots, \rho_K$ are non-negative numbers, then the conic combination $g(x) = \rho_1 f_1(x) + \dots + \rho_K f_K(x)$ is convex.

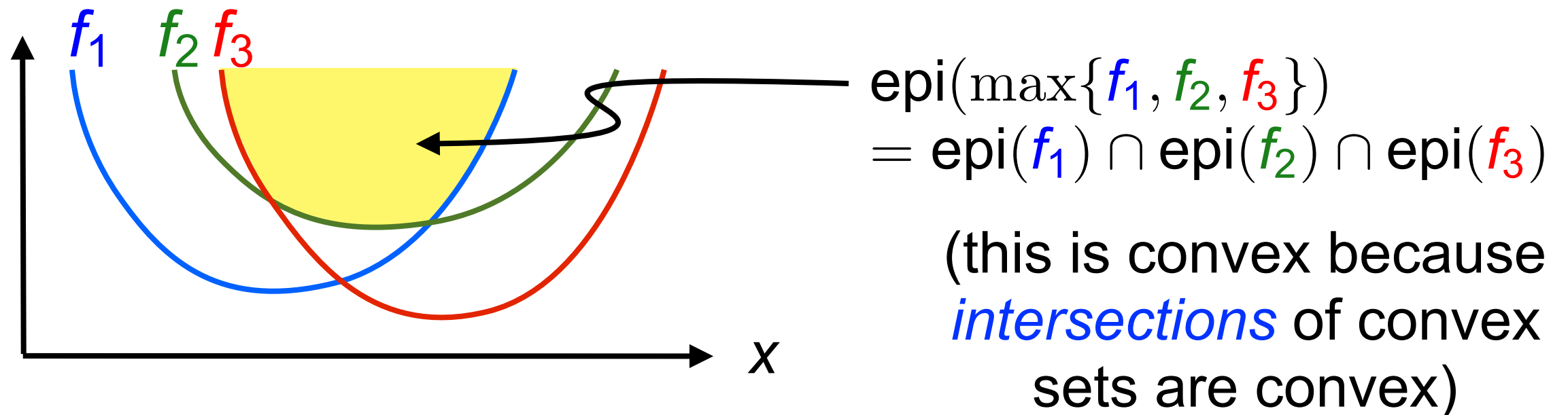
Generalization to *integrals*: If $f(x, y)$ is convex in x for each fixed $y \in \mathcal{Y}$ and $\rho(y)$ is a non-negative function of y , then

$$g(x) = \int_{\mathcal{Y}} \rho(y) f(x, y) dy$$

is convex in x (provided that the integral exists).

Pointwise Maximum and Supremum

Maximum of convex functions: If $f_1, \dots, f_K$ are convex, then the pointwise maximum $g(x) = \max\{f_1(x), \dots, f_K(x)\}$ is also convex.



Supremum of convex functions: If $f(x, y)$ is convex in x for every fixed $y \in \mathcal{Y}$, then the pointwise supremum

$$g(x) = \sup_{y \in \mathcal{Y}} f(x, y)$$

is also convex.

Examples

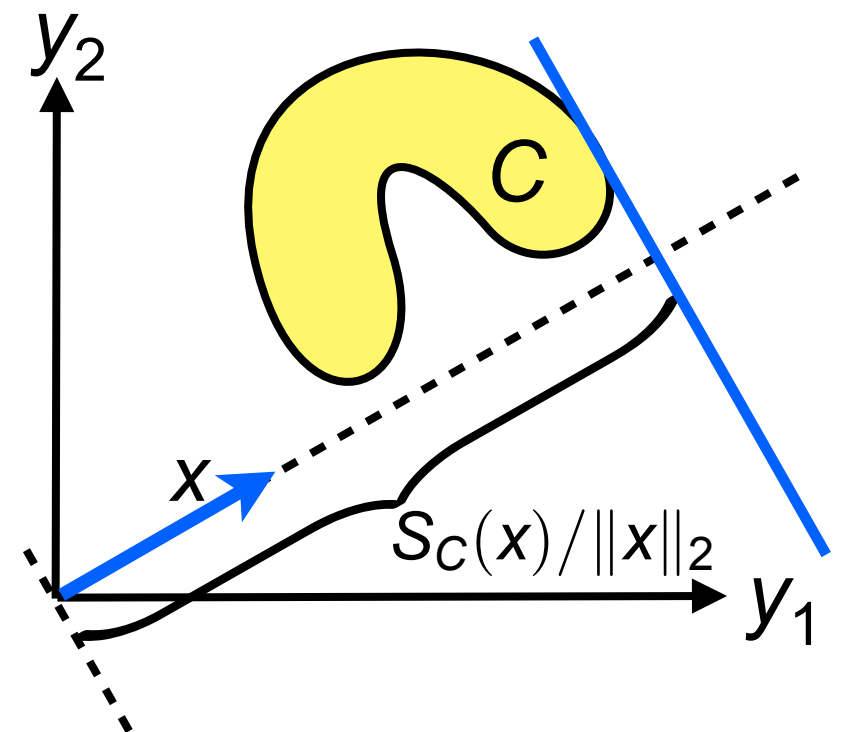
- Piecewise linear functions $f(x) = \max_{k=1,\dots,K} (a_k^\top x + b_k)$ are convex.
- The sum of the r largest components of $x \in \mathbb{R}^n$ is convex as it can be written as a maximum of linear functions.

$$f(x) = \max \{x_{i_1} + x_{i_2} + \dots + x_{i_r} : 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$$

- The support function of a (possibly nonconvex) set C is convex.

$$S_C(x) = \sup_{y \in C} y^\top x$$

The hyperplane $\{y : x^\top y = S_C(x)\}$ is orthogonal to x and “supports” C .



Examples

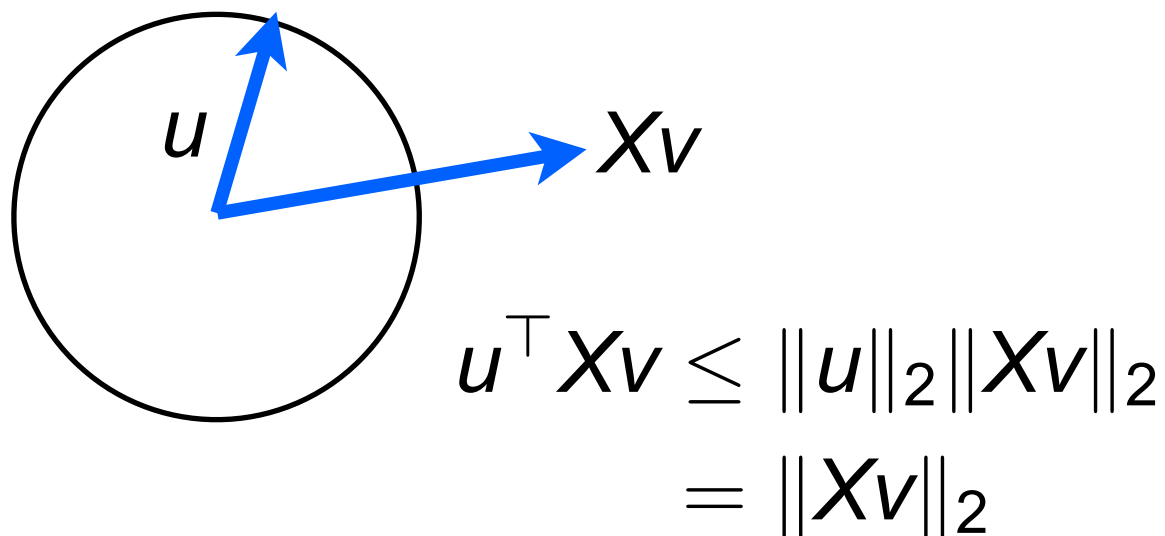
- Maximum eigenvalue $f(X) = \lambda_{\max}(X)$ ($X \in \mathbb{S}^N$)

Write $X = RDR^\top$ with R orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.

$$f(X) = \sup_{\|v\|_2=1} v_1^2 \lambda_1 + \dots + v_n^2 \lambda_n = \sup_{\|v\|_2=1} v^\top D v = \sup_{\|v\|_2=1} v^\top X v$$

- Spectral norm $f(X) = \|X\|_2 = \sup_{v \neq 0} \|Xv\|_2 / \|v\|_2$ ($X \in \mathbb{R}^{m \times n}$)

$$f(X) = \sup_{\|v\|_2=1} \|Xv\|_2 = \sup_{\|v\|_2=1} \sup_{\|u\|_2=1} u^\top X v$$



In both cases, $f(X)$ is the *supremum of linear functions* in X and thus convex.

Composition

Proposition: If $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is *convex* and $h : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* and *non-decreasing*, then $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with $f(x) = h(g(x))$ is convex.

Proof: Choose $x, y \in \mathbb{R}^n$ and $\theta \in [0, 1]$.

$$\begin{aligned}
 f(\theta x + (1 - \theta)y) &= h(g(\theta x + (1 - \theta)y)) && \text{(definition of } f) \\
 &\leq h(\theta g(x) + (1 - \theta)g(y)) && \text{(monotonicity of } h \\
 &&& \text{\& convexity of } g) \\
 &\leq \theta h(g(x)) + (1 - \theta)h(g(y)) && \text{(convexity of } h) \\
 &= \theta f(x) + (1 - \theta)f(y) && \text{(definition of } f)
 \end{aligned}$$

Thus, f is convex.

Example: $\exp(g(x))$ is convex if g is convex

Generalizations

Definition: A function $f : \mathbb{R}^n \rightarrow [-\infty, +\infty)$ is *concave* if $-f$ is convex.

Proposition: If $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is *concave* and $h : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* and *non-increasing*, then $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with $f(x) = h(g(x))$ is convex.

Proposition: If $g : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is *convex* in each component while $h : \mathbb{R}^k \rightarrow \mathbb{R}$ is *non-decreasing* in each argument and *convex*, then $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined via $f(x) = h(g(x))$ is convex.

Proposition: If $g : \mathbb{R}^n \rightarrow \mathbb{R}^k$ is *concave* in each component while $h : \mathbb{R}^k \rightarrow \mathbb{R}$ is *non-increasing* in each argument and *convex*, then $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined via $f(x) = h(g(x))$ is convex.

Minimization

Proposition: If $f(x,y)$ and $g(x,y)$ are convex in (x,y) and C is a convex set, then the optimal value function

$$h(x) = \inf_{y \in C} f(x, y) \quad \text{s.t.} \quad g(x, y) \leq 0$$

is convex.

Proof: Assume that the problem is solvable for every $x \in \text{dom}(h)$. Choose $x_1, x_2 \in \text{dom}(h)$ and let $y_1, y_2 \in C$ be the corresponding minimizers, i.e., $f(x_i, y_i) = h(x_i)$, $g(x_i, y_i) \leq 0$, $i = 1, 2$. For $\theta \in [0, 1]$:

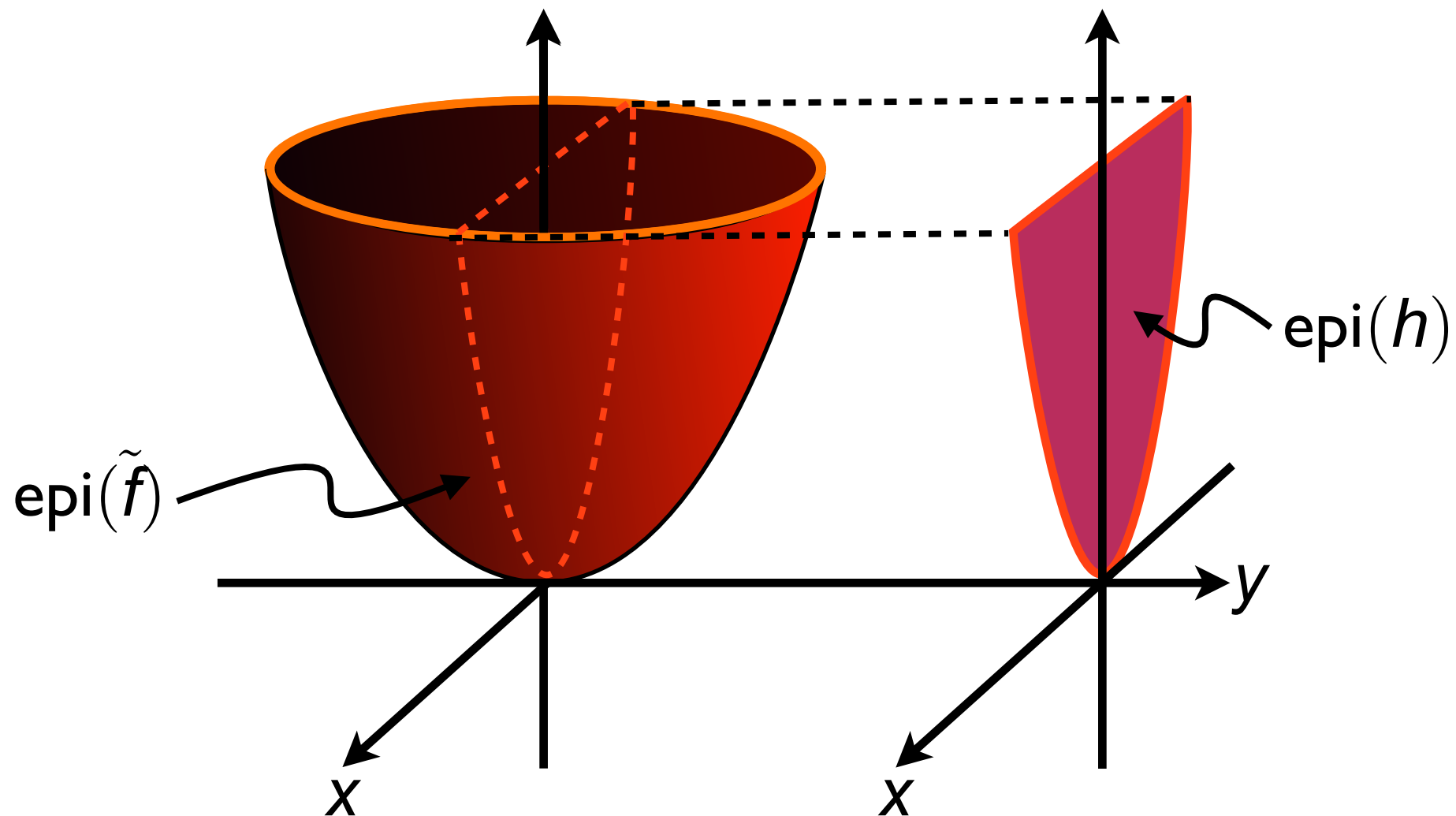
$$\begin{aligned} h(\theta x_1 + (1 - \theta)x_2) &= \inf_{y \in C} \{f(\theta x_1 + (1 - \theta)x_2, y) : g(\theta x_1 + (1 - \theta)x_2, y) \leq 0\} \\ &\leq f(\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2) \\ &\leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2) \\ &= \theta h(x_1) + (1 - \theta)h(x_2) \end{aligned}$$

Thus, h is convex. If the problem is not solvable, one can use a similar argument using ε -optimal solutions for $\varepsilon \downarrow 0$.

Geometric Intuition

We can write $h(x) = \inf_y \tilde{f}(x, y)$, where

$$\tilde{f}(x, y) = \begin{cases} f(x, y) & \text{if } y \in C \text{ and } g(x, y) \leq 0, \\ \infty & \text{otherwise.} \end{cases}$$



If the infimum is attained for every $x \in \text{dom}(h)$, then $\text{epi}(h)$ is the projection of $\text{epi}(\tilde{f})$ along the y -axis. As convexity is preserved under projection, $\text{epi}(h)$ is convex. Thus, h is convex.

Schur's Lemma

Lemma (Schur): Consider $X \in \mathbb{S}^n$ partitioned as $X = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix}$, where $C \succ 0$. Then,

$$X \succeq 0 \iff A - BC^{-1}B^\top \succeq 0.$$

The matrix $A - BC^{-1}B^\top$ is called the *Schur complement* of C .

Proof: Consider the functions $f(x, y) = x^\top Ax + 2x^\top By + y^\top Cy$ and $h(x) = \inf_y f(x, y) = x^\top (A - BC^{-1}B^\top)x$.

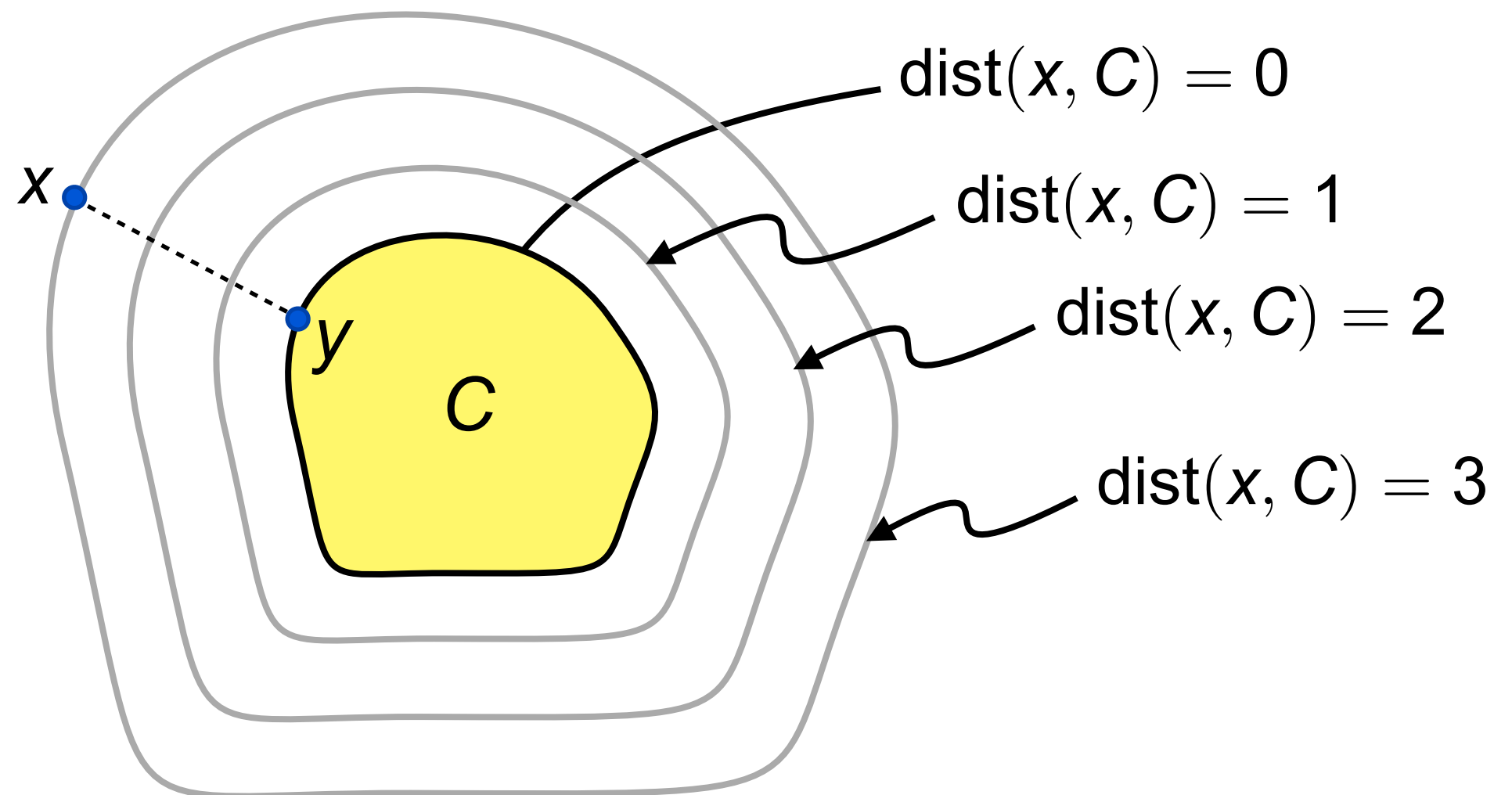
$$\begin{aligned} \implies : \quad X \succeq 0 &\implies f \text{ convex in } (x, y) \implies h \text{ convex in } x \\ &\implies A - BC^{-1}B^\top \succeq 0. \end{aligned}$$

$$\begin{aligned} \impliedby : \quad &\text{We have } A - BC^{-1}B^\top \succeq 0. \text{ Assume } X \not\succeq 0. \\ &\implies \exists (x_0, y_0) \neq 0 \text{ with } f(x_0, y_0) < 0. \\ &\implies h(x_0) = x_0^\top (A - BC^{-1}B^\top)x_0 < 0, \text{ which contradicts} \\ &\text{positive semidefiniteness of the Schur complement. Thus, } X \succeq 0. \end{aligned}$$

Distance Function

The distance of x to a fixed convex set C is convex in x .

$$f(x) = \text{dist}(x, C) = \inf_{y \in C} \|x - y\|_2$$



Perspective Function

Proposition: If $f(x)$ is convex, then the perspective of f , defined as

$$g(x, t) = tf(x/t), \quad \text{dom}(g) = \{(x, t) : (x/t) \in \text{dom}(f), t > 0\},$$

is convex in (x, t) .

Proof: Choose $(x_1, t_1), (x_2, t_2) \in \text{dom}(g)$ and $\theta \in [0, 1]$. Then,

$$\begin{aligned} g(\theta(x_1, t_1) + (1 - \theta)(x_2, t_2)) &= (\theta t_1 + (1 - \theta)t_2) f\left(\frac{\theta x_1 + (1 - \theta)x_2}{\theta t_1 + (1 - \theta)t_2}\right) \\ &= (\theta t_1 + (1 - \theta)t_2) f\left(\frac{\theta t_1 \frac{x_1}{t_1} + (1 - \theta)t_2 \frac{x_2}{t_2}}{\theta t_1 + (1 - \theta)t_2}\right) \\ &\leq \theta t_1 f\left(\frac{x_1}{t_1}\right) + (1 - \theta)t_2 f\left(\frac{x_2}{t_2}\right) \\ &= \theta g(x_1, t_1) + (1 - \theta)g(x_2, t_2) \end{aligned}$$

Thus, g is convex in (x, t) .

Relative Entropy

As the negative logarithm $f(x) = -\log(x)$ is convex on $\mathbb{R}_{++}$, we conclude that its perspective

$$g(x, t) = -t \log(x/t) = t \log(t/x)$$

is convex on $\mathbb{R}_{++}^2$.

The *relative entropy* of two vectors $u, v \in \mathbb{R}_{++}^n$ is defined as

$$\sum_{i=1}^n u_i \log(u_i/v_i).$$

It is convex because it is a sum of n convex functions.

Convexity w.r.t. Generalized Inequalities

Definition: Let $K \subseteq \mathbb{R}^m$ be a proper convex cone. The function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called *K -convex* if

$$f(\theta x + (1 - \theta)y) \preceq_K \theta f(x) + (1 - \theta)f(y) \quad \forall x, y \in \mathbb{R}^n, \theta \in [0, 1].$$

Proposition: If K is a proper convex cone and f is a K -convex function, then the set $C = \{x : f(x) \preceq_K 0\}$ is convex.

Proof: Consider $x, y \in C$ and $\theta \in [0, 1]$. Then,

$$\begin{aligned} f(\theta x + (1 - \theta)y) &\preceq_K \theta f(x) + (1 - \theta)f(y) && (f \text{ is } K\text{-convex}) \\ &\preceq_K 0 && (x, y \in C, K \text{ convex}) \end{aligned}$$

Thus, $\theta x + (1 - \theta)y \in C$, which implies that C is convex.

Example: $f : \mathbb{S}^n \rightarrow \mathbb{S}^n$, $f(X) = X^2$, is $\mathbb{S}_+^n$ -convex.

Main Take-Away Points

- **Definitions:** epigraph, domain and sublevel sets; proper, convex and concave functions;
- **Checking convexity:** using the basic definition; checking convexity along lines; checking the 1st- or 2nd-order conditions (only for differentiable functions)
- **Convexity-preserving transformations:** non-negative weighted sum and integral; composition with affine function; parametric maximum; composition; parametric minimum (check convexity condition!); perspective
- **Schur's lemma:** a block matrix with a positive definite diagonal block is psd if and only if this block's Schur complement is psd
- **Generalized inequalities:** constructing convex sets using K -convex constraint functions and conic inequalities